Verification Plan

SQRT (Sub-zero Quantitative Research Team

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Key Reference Documents

1. ECSS-E-ST-10-02c
2. SQT-CDR-SEP-0
3. SQT-CDR-FTSTP-0
4. UCD-DRD-2026-0

1 Executive Summary

This Verification Plan lays out how we'll prove the SQRT TupperSat mission meets all its defined system and mission requirements. The mission consists of a high-altitude TupperSat platform designed to collect environmental data during ascent and descent and send telemetry to a ground station using an 868 MHz radio link.

We'll verify each part (components, subsystems, and the whole system) using a mix of tests, analysis, inspections, and design reviews. A prioritized element to test is the communication system. We're especially focused on making sure the communications work reliably over the long distances we expect for the maximum expected mission duration.

A Verification Matrix links each requirement to its corresponding verification method and verification level, ensuring full traceability. Once this process is done, the mission will be ready to fly and meets both the mission-specific and inherited UCD requirements.

2 Verification Subject

Mission Statement: The SQRT mission will produce and observe the flash evaporation of water via exposure to a low pressure and low temperature environment to investigate how such plumes thermally evolve.

This Verification Plan covers the plan for verifying requirements for the SQRT mission designed to investigate the mission statement above. This investigation into the thermal mechanisms behind cryovolcanic plumes and water desalinisation processes will be done using a high-altitude demonstration of a payload. Mission success depends on verifying the TupperSat platform, payload, onboard sensors, a power system, and an 868 MHz radio link that sends telemetry back to the ground.

The plan involves verifying how the entire system works together at a component and subsystem level. It lays out the steps our team will follow to show that the system meets both the inherited UCD requirements and the specific technical goals set out in SQT-CDR-SEP-0 and SQT-PDR-MDD-0 at the Critical Design Review. The aim of the verification plan is to ensure the integrated system is qualified and ready to fly.

3 Verification Approach

3.1 General Approach

The verification approach for the SQRT TupperSat mission is based on demonstrating compliance with all Level 1 (mission) and Level 2 (technical) requirements defined in the System Engineering Plan. Verification activities are structured hierarchically and performed progressively: first at the component level, then at the subsystem level, and finally at the full system level.

Each stage builds on the last. When individual components and subsystems pass their tests, it gives confidence for the bigger system checks ahead. The final round of testing looks at the whole system in conditions that match real operations, ensuring readiness for flight acceptance.

3.2 Verification Methods

In accordance with standard systems engineering practice, four verification methods will be employed:

Verification by Test (T) involves direct experimental validation under controlled conditions: such as running communication range tests, checking how long the satellite can operate, validating sensors, and ensuring everything works together as a system.

Verification by Analysis (A) uses calculations and models to check things like power use, link budgets, and how long the mission will last.

Verification by Inspection (I) consists of visually checking or measuring mass, structure, and hardware configuration.

Verification by Review-of-Design (R) involves confirmation that the design documents themselves prove everything is in order, covering software architecture, interfaces, and whether the paperwork lines up with reality.

Every requirement in the Verification Matrix gets at least one of these verification methods.

3.3 Verification Levels

Verification activities are organised into three hierarchical levels:

Component Level Verification checking that hardware parts and software modules do what they're supposed to before anything gets put together.

Subsystem Level Verification make sure groups of components, like the communications, power, or payload subsystem, work as intended.

System Level Verification demonstrates that the fully integrated TupperSat system satisfies overall mission and technical requirements, including full operational tests such as long-range communications and endurance trials.

Success at each level depends on passing the test before it. This structure keeps the process transparent and ensures every requirement is fully met.

4 Verification Control Methodology

Verification of the SQRT TupperSat system will be controlled and tracked through a structured Verification Matrix, which forms the core of the Verification Control Document (VCD) and links every requirement from the System Engineering Plan to its specific verification method and level.

At Critical Design Review, each requirement is assigned:

- a unique verification method (Test, Analysis, Inspection, or Review-of-Design),
- a defined verification level (component, subsystem, or system),
- and a reference to the planned test or verification activity.

During Phase 3 (Production and Qualification), the verification campaign will be executed according to the defined plan. After every test is done, we create a Test Specification and Test Plan (TSTP) and a Test Report (TRPT). We then update the Verification Matrix with these documents and mark the compliance status.

If something doesn't meet requirements or there's an anomaly, we document what went wrong, take corrective action, and do another round of verification if needed. We only close out a requirement once we've reviewed and accepted solid proof of compliance.

When everything is done, the Verification Matrix, now complete with compliance status and all supporting test documentation, forms the foundation of the Verification Control Document that we submit at the Flight Acceptance Review.

5 Verification Matrix

5.1 Level 1: Mission Requirements

5.1.1 Customer (UCD) Mission Requirements

Table 1: Verification Matrix: Customer (UCD) Mission Requirements

Requirement ID/Title	Verification Level	Verification Method	Verification Reference	Pass/Fail Criteria	Status	Notes
UCD:MR:0010 Payload Size	System	Review-of-Design + Inspection	SQT-CDR-HDF-0	Dimensions and mass comply with UCD PRD constraints	Planned	Mechanical design + mass & dimension check
UCD:MR:0020 Flight Time	System	Test + Analysis	SQT-CDR-FTSTP-0	System operates ≥6 hours continuously	Planned	Verified during endurance test
UCD:MR:0030 Flight Environment	System	Test + Analysis	SQT-CDR-HDF-0 + SQT-CDR-FTSTP-0	All subsystems operate within specified environmental limits	Planned	Compliance demonstrated through component qualification and integration test
UCD:MR:0040 TDRS Communication	System + Subsystem	Test + Analysis	SQT-CDR-FTSTP-0	Successful bidirectional communication with ground station	Planned	Verified via long-range communication test
UCD:MR:0050 Tracking and Recovery	System	Test	SQT-CDR-FTSTP-0	Real-time position telemetry received at ground station	Planned	GPS + telemetry verification
UCD:MR:0060 Non-recovery Data Transmission	System	Test	SQT-CDR-FTSTP-0	Experimental data transmitted successfully during flight simulation	Planned	Ensures data availability if recovery fails

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Requirement ID/Tile	Verification Level	Verification Method	Verification Reference	Pass/Fail Criteria	Status	Notes
UCD:MR:0070 Athmospheric profiling	System	Test + Analysis	SQT-CDR-FTSTP-0	Measurement resolution ≥1 per 200m ascent	Planned	Verified via sensor sampling rate and altitude correlation
UCD:MR:0080 Launch Readiness	System	Test + Inspection	SQT-CDR-FTSTP-0 will become OUM in FAR	System operational without further intervention at delivery	Planned	Verified during final system integration test

5.1.2 Mission Specific Requirements

Table 2: Verification Matrix: Mission Specific Requirements

Requirement ID/Title	Verification Level	Verification Method	Verificatio Reference	Pass/Fail Criteria	Status	Notes
SQT:MR:010 Thermal Data	System + Subsystem	Test + Analysis	SQT-CDR-FTSTP-0 + SQT-CDR-SDF-0	Temperature sensor accuracy within specified tolerance (e.g. ±0.5°C) and data successfully recorded	Planned	Verified via sensor calibration and integrated data acquisition test
SQT:MR:030 Timestamped Data	Subsystem	Test + Review-of-Design	SQT-CDR-SDF-0 + SQT-CDR-FTSTP-0	Timestamp recorded for each primary data sample	Planned	Verified via software validation
SQT:MR:040 Altitude Data	System + Subsystem	Test	SQT-CDR-FTSTP-0 + SQT-CDR-SDF-0	Altitude recorded and logged with each data acquisition event	Planned	Verified during integration testing
SQT:MR:050 Trigger Operation	System + Subsystem	Test	SQT-CDR-FTSTP-0	System confirms trigger condition met and logs confirmation	Planned	Critical for mission event validation

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Requirement ID/Tile	Verification Level	Verification Method	Verificatio Refer-ence	Pass/Fail Criteria	Status	Notes
SQT:MR:060 Exposed Sample Containment	Subsystem	Test + Inspection	SQT-CDR-FTSTP-0 + SQT-CDR-HDF-0	Water sample remains contained for defined exposure duration	Planned	Verified during grounf-base exposure simulation
SQT:MR:070 External Environment Readings	Subsystem	Test	SQT-CDR-FTSTP-0	Continuous pressure data recorded during exposure	Planned	Used for post-flight comparison
SQT:MR:080 Ease of Data Retrieval	System	Test + Review-of-Design	SQT-CDR-SDF-0 + SQT-CDR-FTSTP-0	All primary data retrievable after recovery without corruption	Planned	Linked to DMP prodecures
SQT:MR:090 Viability for Prototyping	System	Test + Review-of-Design	SQT-CDR-HDF-0 + SQT-CDR-FTSTP-0	System supports repeatable ground testing without redesign	Planned	Demonstrated during Phase 2 testing
SQT:MR:100 Water Exposure	System	Test + Analysis	SQT-CDR-FTSTP-0 + SQT-CDR-HDF-0	Exposure occurs at pressure $\leq$ 35 mbar equivalent	Planned	Verified via a thermal sensor detecting if the water has passed through the valve

5.2 Level 2: Technical/System Requirements

5.2.1 Customer (UCD) Mission Requirements

5.2.1.1 Structure

Table 3: Verification Matrix: Structural Requirements

Requirement ID/Title	Verification Level	Verification Method	Verification Reference	Pass/Fail Criteria	Status	Notes
UCD:TR:00010 TupperSat Mass	System + Component	Review-of-Design + Inspection	SQT-CDR-HDF-0	Total mass $\leq$ 800g	Planned	Verified before final integration
UCD:TR:00020 TupperSat Volume	System	Inspection + Review-of-Design	SQT-CDR-HDF-0	Enclosed volume $<$ 2 litres	Planned	Verified during integration
UCD:TR:00030 Launch Interface	System + Subsystem	Test + Inspection	SQT-CDR-HDF-0 + SQT-CDR-FTSTP-0	Verificaion by limited stress test and FEA (Final Element Analysis)	Planned	Verified during final assembly

5.2.1.2 Communications

Table 4: Verification Matrix: Communications Requirements

Requirement ID/Title	Verification Level	Verification Method	Verification Reference	Pass/Fail Criteria	Status	Notes
UCD:TR:00100 Packet Transmission Frequency (868MHz)	Subsystem	Test	SQT-CDR-FTSTP-0	Transmission frequency = 868 MHz	Planned	Verified during communication subsystem test
UCD:TR:00110 Packet drop rate $<$ 20% over $\geq$ 12 km	System	Test+ Analysis	SQT-CDR-FTSTP-0	Packet drop rate $<$ 20% over $\geq$ 12 km	Planned	Verified via ground test
UCD:TR:00120 Telemetry Packet Rate	Subsystem	Test	SQT-CDR-FTSTP-0	2-3 housekeeping packets per minute	Planned	Verified via ground test
UCD:TR:00130 Data Packet Rate	Subsystem	Test	SQT-CDR-FTSTP-0	$\geq$ 3 data packets per minutes	Planned	Verified during ground test

5.2.1.3 Housekeeping Telemetry and Atmospheric Data

Table 5: Verification Matrix: Housekeeping Telemetry and Athmospheric Data

Requirement ID/Title	Verification Level	Verification Method	Verification Reference	Pass/Fail Criteria	Status	Notes
UCD:TR:00200 Telemetry Altitude	Subsystem	Test	SQT-CDR-FTSTP-0 + SQT-CDR-SDF-0	Altitude updated every 10 s, accuracy ±50 m	Planned	Verified during integration
UCD:TR:00210 Telemetry Lat/Long	Subsystem	Test	SQT-CDR-FTSTP-0 + SQT-CDR-SDF-0	Horizontal accuracy ±50 m	Planned	Verified during integration
UCD:TR:00220 Internal Temperature	Subsystem	Test + Calibration	SQT-CDR-FTSTP-0 + SQT-CDR-HDF-0	Accuracy ±2°C, sampling ≥ 20 s	Planned	Verified during integration
UCD:TR:00230 External Temperature	Subsystem	Test + Calibration	SQT-CDR-FTSTP-0 + SQT-CDR-HDF-0	Accuracy ±2°C, sampling ≥ 20 s	Planned	Verified during integration
UCD:TR:00240 Athmospheric Pressure	Subsystem	Test + Calibration	SQT-CDR-FTSTP-0 + SQT-CDR-HDF-0	Accuracy ±20mbar, sampling ≥ 20 s	Planned	Verified during integration

5.2.1.4 Power

Table 6: Verification Matrix: Power Requirements

Requirement ID/Title	Verification Level	Verification Method	Verification Reference	Pass/Fail Criteria	Status	Notes
UCD:TR:00300 TupperSat Power	System	Test + Analysis	SQT-CDR-FTSTP-0 + SQT-CDR-HDF-0	System operate ≤ 6 hours continuously	Planned	Verified via 6hours bench test

5.2.1.5 Operations

Table 7: Verification Matrix: Operations Requirements

Requirement ID/Title	Verification Level	Verification Method	Verification Reference	Pass/Fail Criteria	Status	Notes
UCD:TR:00400 Delivery	System	Test + Inspection	SQT-CDR-FTSTP-0	System ready for launch within 15 minuted	Planned	Verified after integration by a third party with our Launch Operation Manuel

5.2.2 Mission Specific Requirements

5.2.2.1 Water Container

Table 8: Verification Matrix: Water Container Requirements

Requirement ID/Title	Verification Level	Verification Method	Verification Reference	Pass/Fail Criteria	Status	Notes
SQT:TR:00100 Water Temperature (> 25°C before exposure)	Subsystem	Test	SQT-CDR-FTSTP-0 + SQT-CDR-HDF-0	Temperature ≥ 25°C before exposure	Planned	Verified during integration, the water container shall be well isolated
SQT:TR:00110 Pressure Resistant	Subsystem	Test + Inspection	SQT-CDR-HDF-0 + SQT-CDR-FTSTP-0	No structural damage or leakage during ascent simulation	Planned	Verified during Phase 2 testing

5.2.2.2 Exposure Mechanism

Table 9: Verification Matrix: Exposure Mechanism Requirements

Requirement ID/Title	Verification Level	Verification Method	Verification Reference	Pass/Fail Criteria	Status	Notes
SQT:TR:00200 Valve Activation	System + Subsystem	Test	SQT-CDR-FTSTP-0	Valve activation signal logged and transmitted	Planned	Verified during ground test
SQT:TR:00210 Leak Avoidance	Subsystem	Test	SQT-CDR-FTSTP-0 + SQT-CDR-HDF-0	No premature leak before trigger altitude and pressure	Planned	Verified during ground test
SQT:TR:00220 Valve Operation (-40°C)	Subsystem	Test	SQT-CDR-HDF-0 + SQT-CDR-FTSTP-0	Valve functinal at -40°C	Planned	Verified during ground test
SQT:TR:00230 Tubing material (-40°C)	Component	Test + Inspection	SQT-CDR-HDF-0	Tubing remains operational at -40°C	Planned	Verified during ground test
SQT:TR:00240 Telemetry - Atmospheric Pressure	Subsystem	Test + Analysis	SQT-CDR-HDF-0	Recording of athmopheric pressure with accuracy better than 20mbar at sea level	Planned	Verified during ground test

5.2.2.3 Thermal Sensor

Table 10: Verification Matrix: Thermal Sensor Requirements

Requirement ID/Title	Verification Level	Verification Method	Verification Reference	Pass/Fail Criteria	Status	Notes
SQT:TR:00300 Thermal Resolution (≤0.1°C)	Subsystem	Test	SQT-CDR-HDF-0 + Thermal Sensor Datasheet (MLX90640) + SQT-CDR-FTSTP-0	Resolution ≤0.1°C confirmed	Planned	Verified during integration
SQT:TR:00310 Sensor Function (-10°C)	Subsystem	Test	Thermal Sensor Datasheet (MLX90640) + SQT-CDR-FTSTP-0	Camera operational at −10°C	Planned	Verified in the data sheet of the thermal sensor (MLX90640) (-40°C)
SQT:TR:00320 Frame Rate (>30 fps)	Subsystem	Test	Thermal Sensor Datasheet (MLX90640) + SQT-CDR-FTSTP-0	Frame rate ≥ 30 fps	Planned	Verified in the data sheet of the thermal sensor (MLX90640)
SQT:TR:00330 Calibration Stability	Subsystem	Test + Analysis	Thermal Sensor Datasheet (MLX90640) + SQT-CDR-FTSTP-0	Calibration maintained over expected flight range	Planned	Verified in the data sheet of the thermal sensor (MLX90640)
SQT:TR:00340 Field of View (≥ 15°)	Subsystem	Test + Inspection	Thermal Sensor Datasheet (MLX90640) + SQT-CDR-HDF-0	Field of View of≥ 15° confirmed	Planned	Verified in the data sheet of the thermal sensor (MLX90640)

5.2.2.4 Secondary Data Acquisition

Table 11: Verification Matrix: Sensors Requirements

Requirement ID/Title	Verification Level	Verification Method	Verification Reference	Pass/Fail Criteria	Status	Notes
SQT:TR:00420 Temperature Sensor Sampling Rate	Subsystem	Test	SQT-CDR-FTSTP-0 + SQT-CDR-SDF-0	Sampling rate sufficient to capture temperature peaks and water passage through the tubing	Planned	Verified after integration

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Requirement ID/Tile	Verification Level	Verification Method	Verification Reference	Pass/Fail Criteria	Status	Notes
SQT:TR:00500 Altitude Sensor Update Rate	Subsystem	Test	SQT-CDR-FTSTP-0 + SQT-CDR-SDF-0	Altitude updated every 10 s	Planned	Verified during ground test

5.2.2.5 Communications

Table 12: Verification Matrix: Mission Specific Communications Requirements

Requirement ID/Title	Verification Level	Verification Method	Verification Reference	Pass/Fail Criteria	Status	Notes
SQT:TR:00600 Downlink Capacity	System	Test	SQT-CDR-FTSTP-0 + SQT-CDR-SDF-0	Communications architecture capable of transmitting all required data packets to ensure the ecquisition of mission validation figures	Planned	Verified during ground test

5.2.2.6 Triggering Algorithms

Table 13: Verification Matrix: Triggering Algorithms Requirements

Requirement ID/Title	Verification Level	Verification Method	Verification Reference	Pass/Fail Criteria	Status	Notes
SQT:TR:00700 Pressure Trigger	System + Subsystem	Test + Review-of-Design	SQT-CDR-SDF-0 + SQT-CDR-FTSTP-0	Trigger activates when measured pressure $\leq$35 mbar	Planned	Verified via bench test using simulated telemetry data
SQT:TR:00710 Altitude Trigger	Subsystem	Test + Analysis	SQT-CDR-SDF-0 + SQT-CDR-FTSTP-0	Trigger activates at altitude $\geq$ 23 km when pressure data flagged inconsistent	Planned	Verified via bench test using simulated telemetry data
SQT:TR:00720 Falling Trigger Operation	System + Subsystem	Test	SQT-CDR-SDF-0 + SQT-CDR-FTSTP-0	Trigger activates if descent detected continuously for $\geq$10 s	Planned	Verified via bench test using simulated telemetry data

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Requirement ID/Tile	Verification Level	Verification Method	Verification Reference	Pass/Fail Criteria	Status	Notes
SQT:TR:00730 Falling Trigger Minimum Altitude	System	Test	SQT-CDR-FTSTP-0 + SQT-CDR-SDF-0	Falling trigger does not activate below 2 km altitude	Planned	Verified via bench test using simulated telemetry data

This Verification Matrix gives tracebility between mission and technical requirements and the planned verification plan. The team will use this matrix to support the Verification Control Document to complete the Flight Acceptance Review (FAR).